School of Computing & Information Sciences

Spring 2026 Senior Design Project

FIU Engineering & Computing
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Market Sentiment & Momentum: Evaluating Predictive Signals across the Magnificent Seven

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Problem & Motivation

- The "Magnificent Seven" ***(AAPL, AMZN, MSFT, NVDA, GOOGL, META, TSLA)*** have historically moved together but show increasing fragmentation.
- Despite enormous volume of daily news coverage surrounding these firms, it remains unclear whether public sentiment expressed in financial news carries any signal with their short-term price behavior.
- **Goal:** Build reproducible pipeline to test whether the sentiment of public news coverage is associated with next-day price behavior in MAG7 technology stocks.

Workflow Pipeline

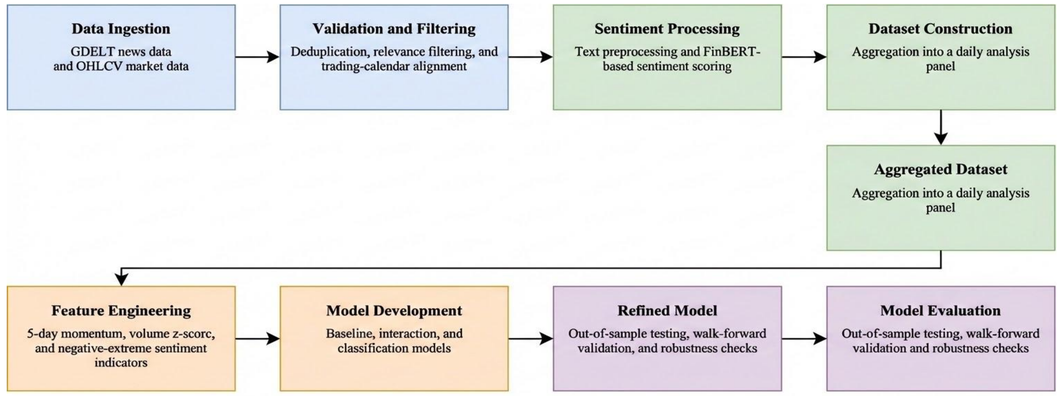
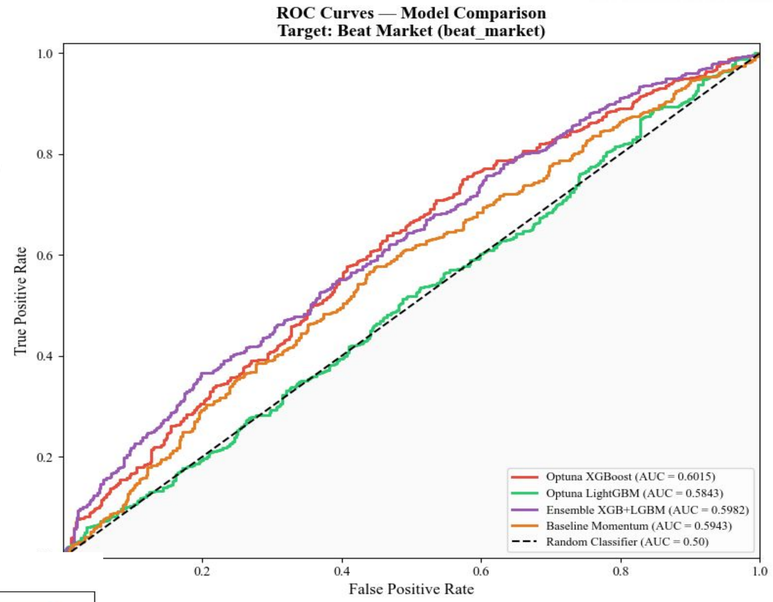


Figure 1. Research pipeline for sentiment-based analysis of MAG7 stocks.

Hypotheses Testing

- **H1**: Sentiment score correlates with next-day returns → ***Rejected***
- **H2**: Sentiment a lagging indicator? *news reacts to prices* → ***Rejected***
- **H3**: Extreme negative sentiment elevates next-day downside risk → ***Rejected***
- **H4**: Sentiment moderates the momentum-return relationship → ***Rejected***
- **H5**: Walk-forward validation confirms H4's stability → ***Rejected***
- **H6**: Sentiment acts as a regime-dependent momentum modifier → ***Inconclusive***

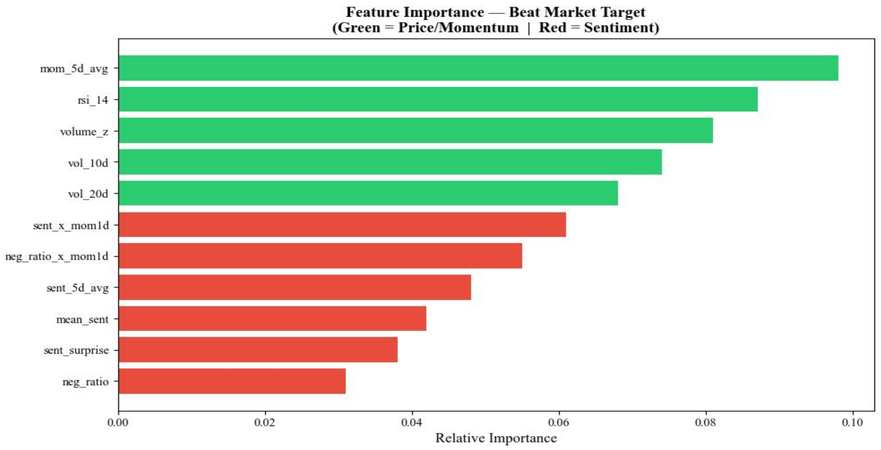
Verification



Key Findings

- **Momentum** emerged as the dominant signal in the analysis. Among all tested features, 5-day momentum was the strongest and most consistently reliable predictor.
  Spearman correlation of $\rho = +0.167$ ($p < 0.0001$) with the probability of beating the market.
- **Sentiment alone** did not demonstrate meaningful predictive power. Its relationship with next-day price behavior was weak and unstable across specifications.
- Extreme negative sentiment showed limited in-sample interaction effects with momentum, but these relationships did not remain stable under out-of-sample evaluation.
  No robust standalone predictive claim can be made for sentiment-based measures in this study.

Implementation



Summary

- Across all **6** hypotheses, sentiment alone showed *no statistically significant* predictive power.
- Momentum emerged as the dominant signal and our best model,
  - Optuna XGBoost, achieved AUC = **0.6015** *approximately 5 percentage points above random*.
- For heavily covered stocks like the MAG 7, aggregated news sentiment analysis is *insufficient* for predicting price movement.

Acknowledgement

The material presented in this poster is based upon the work supported by Dr. Masoud Sadjadi.. We/I thank Dr. Masoud Sadjadi for assistance, cooperation and mentorship that I/we received throughout this process